MAE5005 Computational Fluid Dynamics, Fall 2026

Homework Set 4

Due Wednesday, April 1, 2026 by 23:59, by online submission through *Blackboard*

12. (30 points) Consider again the Problem 11, and the central-differencing-in-space and forward-Euler-in-time finite-difference formulation is

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = g + \nu \frac{u_{j-1}^n - 2u_j^n + u_{j+1}^n}{\Delta y^2}, \quad \text{for} \quad j = 2, 3, 4, ..., N-1, N. \tag{1}$$

In Problem 11, we have used the following notations: u_j^n represents the actual numerical solution from the code, $u_m(y,t)$ is the analytical solution for the modified equation, and $u(y,t)$ is the analytical solution of the original partial differential equation.

(a) For $N = 8$ (A coarse mesh) and $\alpha \equiv \nu \Delta t / \Delta y^2 = 0.32$ as used in Problem 11, compute and plot the numerical solution u_j^n and the exact solution of the finite-difference scheme

$$\frac{u_{j,theory}^n(y,t)}{u_0} = \left(1 - \frac{y_j^2}{L^2}\right) - \sum_{k=0}^{k_{max}} \frac{4(-1)^k}{[\pi(k+0.5)]^3} \left[1 - 2\alpha + 2\alpha \cos(\beta_k \Delta y)\right]^n \cos\left[\beta_k y_j\right] \tag{2}$$

at the location $y = 0.5L$ as a function of time. Here $\beta_k = (k + 0.5)\pi/L$. Note here that k_{max} is also an important parameter.

(b) For part (a) at the same location $y = 0.5L$, plot (i) the normalized difference $\left(u_j^n - u_{j,theory}^n\right)/u_0$, (ii) $\left(u - u_{j,theory}^n\right)/u_0$, (ii) $\left(u_m - u_{j,theory}^n\right)/u_0$, as a function of time. Comment and interpret your results. Also study how your results change with your choice of k_{max}.

(c) Repeat (a) and (b) for $\alpha \equiv \nu \Delta t / \Delta y^2 = 0.53$. Comment on your results.

13. (40 points) Consider the 1D advection equation (with $a = 1$)

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0, \quad \text{for } 0 \le x \le 2 \text{ and } t \ge 0, \tag{3}$$

with the following initial condition and periodic boundary condition

$$u(x, t = 0) = \sin(\pi x) + 0.6 \sin(3\pi x), \quad u(x = 0, t) = u(x = 2, t). \tag{4}$$

Namely, the initial condition contains two modes with wavenumbers $k_1 = 1$ and $k_2 = 3$.

Consider the following three explicit schemes for solving the above advection equation

$$\text{Upwind}: \quad \frac{u_j^{n+1} - u_j^n}{\Delta t} + a \frac{u_j^n - u_{j-1}^n}{\Delta x} = 0, \tag{5}$$

$$\text{Downwind}: \quad \frac{u_j^{n+1} - u_j^n}{\Delta t} + a \frac{u_{j+1}^n - u_j^n}{\Delta x} = 0, \tag{6}$$

$$\text{Central}: \qquad \frac{u_j^{n+1} - u_j^n}{\Delta t} + a\frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = 0, \tag{7}$$

Assume N grid cells are used, namely, $\Delta x = 2/N$ and $C \equiv a\Delta t/\Delta x = 0.5$. Also, let the node points are located at $x_j = j\Delta x$, for $j = 0, 2, .., N-1$. Hint: For easy coding, you may add one or two virtual nodes on each end of the domain, as needed. The periodic boundary condition implies that $u_j = u_{j+N}$.

(a) Find the exact solution $u(x,t)$ for the system defined by equations (3) and (4).

(b) Use $N = 20$, plot the numerical solutions for the three schemes and the analytical solution of Eq. (3), together, as a function of x, at $t = 0.5$ and $t = 1.5$, respectively.

(c) Use $N = 160$, plot the numerical solutions for the three schemes and the analytical solution of Eq. (3), together, as a function of x, at $t = 0.5$ and $t = 1.5$, respectively.

(d) For $N = 160$, study how the L1 and L2 errors from the three schemes vary with time. Here the L1 and L2 errors are defined as

$$L_1 \equiv \frac{1}{N}\sum_{n=0}^{j=N-1} \left|u_j^n - u(x_j, t_n)\right|, \qquad L_2 \equiv \sqrt{\frac{1}{N}\sum_{n=0}^{j=N-1} \left(u_j^n - u(x_j, t_n)\right)^2}. \tag{8}$$